Audit of the SV efficiency proof, round 6: after the T-corrections

Source audited: `sv_efficiency_proof.tex` (with `sv_efficiency_proof.md` checked for sync), at the commit "Fix audit round-5 gap T5 and minor issues". Earlier rounds: 1, 2, 3, 4, 5. Not refereed.

Part A. Proof-check

A1. Scope

This round checks:

- the passages changed in round 5 (Theorem 4 with $c_{b,\sigma}$, Remark 9.3, Lemma L, Remarks 5.3 and 5.4, §0, §7, Lemma A step 2);
- the derivations not re-derived in earlier rounds: the squeezed-bubble scalings of Tool A3, the face step of Lemma D, Lemma G case 2 step 4, and the coercivity behind Remark 1.2;
- the sync between `.tex` and `.md` (all audit markers match; one apparent mismatch is a line break inside "audit R1" in the `.tex`).

A2. Step table

step	tool	hypothesis	constant	status
Rem. 1.2, coercivity	$B(e,e)=\varepsilon\|\nabla e\|^2+((c-\frac12\mathrm{div}\,b)e,e)+\frac12\int_{\Gamma_N}b\cdot n\,e^2$	H3, H4 ($b\cdot n\ge0$ on Γ_N), H5 ($e\in H^1_D$)	1	OK, both entries of c_b
Tool A3, face bubble	$\|b_F^\theta\mu\|_{\omega_F}\lesssim\theta^{1/2}h^{1/2}\|\mu\|_F=\varepsilon^{1/4}\alpha_F^{1/2}\|\mu\|_F$; $\varepsilon^{1/2}\|\nabla b_F^\theta\mu\|\lesssim\varepsilon^{1/2}\theta^{-1/2}h^{-1/2}\|\mu\|_F=\varepsilon^{1/4}\alpha_F^{-1/2}\|\mu\|_F$; $\sigma^{1/2}\|b_F^\theta\mu\|\le\varepsilon^{1/4}\alpha_F^{-1/2}\|\mu\|_F$ since $\sigma\alpha_F^2\le1$	$\theta_F=\varepsilon^{1/2}\alpha_F/h_F\le1$ (so the min in A3 is redundant)	$C(d,\vartheta)$	OK (re-derived)
Lemma D, face	$c\|j_F\|^2\le\ell\|_{,\omega_F}\tnorm{b\mu}+\|r_\ell\|_{\omega_F}\|b\mu\|$ ⇒ $\varepsilon^{-1/4}\alpha_F^{1/2}\|j_F\|\lesssim\|\ell\|_{,\omega_F}+\alpha_F\|r_\ell\|_{\omega_F}$	$b_F^\theta\mu\in H^1_*(\omega_F)$ (G1)	$C(d,\vartheta)$	OK; matches the face weight $\varepsilon^{-1/2}\alpha_F\|j_F\|^2$ in RES
Lemma G case 2, step 4	Poincaré/Friedrichs on $H^1_\dagger(\omega_a)$; trace $\|w\|_F\lesssim h^{1/2}\|\nabla w\|$	A2's test space (T1)	$C(d,\vartheta)$	OK
Lemma G case 2, step 6	$\theta\ge1$ ⇒ $\alpha_K=h\varepsilon^{-1/2}$, $\varepsilon^{-1/2}\alpha_F=\varepsilon^{-1}h$	—	exact	OK
Thm 4.2–4.3 with $c_{b,\sigma}$	local bound via $\|e\|_{\omega_a}\le\sigma^{-1/2}\tnorm{e}_{\omega_a}$	σ > 0, or $c=\frac12\mathrm{div}\,b$ (which forces σ = 0 by H3)	$C(1+c_{b,\sigma})$, $C(1+c_{b,\sigma})^2$	OK (T5 closed)
Rem. 9.3, local Friedrichs	$\|v\|_{\omega_a}\le C_{Ph,a}\|\nabla v\|_{\omega_a}$ for $v\in H^1_*(\omega_a)$; σ = 0 ⇒ $\tnorm{v}_{\omega_a}=\varepsilon^{1/2}\|\nabla v\|_{\omega_a}$	$\partial\omega_a\setminus\Gamma_N$ of measure $\simeq h_a^{d-1}$	$C_P(d,\vartheta)$	OK

step	tool	hypothesis	constant	status
Rem. 9.3, parts 2–3	as in Theorem 4, with μ_a; part 3's first step is global, so κ with the Friedrichs-based c_b is admissible there	—	C, $C\kappa$	OK
Rem. 9.3, last sentence	"the extra term is of higher order where $h_a\ll \varepsilon^{1/2}$"	—	—	**U1**: wrong scaling
Lemma L (scope), Remarks 5.3/5.4, §0, §7, Lemma A step 2	—	—	—	OK (T2–T4, T6, T8 closed)

A3. Verdict

- **All results are correct**, given A2 (Neumann version, [EV15]) and A5 (Licht 2019, abstract-level).
- **U1** is one wrong side claim in Remark 9.3. It affects no statement and no computed bound.

A4. U1 and its fix

- **The claim.** Remark 9.3 calls $\sum_a\mu_a^2$, with $\mu_a=C_P|c-\frac12\mathrm{div}\,b|_{\infty,\omega_a}h_a\varepsilon^{-1/2}|e|_{\omega_a}$, a term "of higher order where $h_a\ll\varepsilon^{1/2}$".
- **The correct bound.** By the global Friedrichs inequality on H^1_D ($e\in H^1_D$ by H5),

$$\Bigl(\sum_a\mu_a^2\Bigr)^{1/2}\le\sqrt{d+1}\,C_PC_F\,|c-\tfrac12\mathrm{div}\,b|_\infty\,\frac{h_{\max}}{\varepsilon}\,\tnorm{e}.$$ So μ is dominated by $\tnorm{e}$ when $h\lesssim\varepsilon/(C_PC_F|c-\frac12\mathrm{div}\,b|_\infty)$, not when $h\ll\varepsilon^{1/2}$.

- **Consequence.** In the convection-dominated regime $h\gg\varepsilon$, the extended local measure controls a genuinely stronger quantity (a weighted L^2 error). For σ = 0 with $c\ne\frac12\mathrm{div}\,b$, Theorem 4.2 is therefore a weaker efficiency statement there.
- **Fix.** Replace the sentence by the bound above and its consequence. The sharper remedy is the cellwise-σ track: it puts $c-\frac12\mathrm{div}\,b$ into the norm, so this case needs no extra term at all.

A5. Optimality (changes since round 5)

- **The cellwise-σ track** (adapting [AV14]) now removes three restrictions at once: the c_b factors of Corollaries 3/3', the σ-based restriction of Theorem 4.2–4.3 (T5), and the extra term μ of Remark 9.3 (U1).
- **Otherwise unchanged.** The hypotheses-used list is that of round 5, with the Friedrichs-based c_b admissible in the global results only.

Part B. Literature scan (round 6)

B1. Answer

- **No change in novelty or track.**
- **This round re-tested the main novelty claim of §9:** local efficiency in a localisable augmented norm, with the stabilisation's nodal moments as the only global term. The closest results still stop short of it:
  - **ESV10** (DG): local lower bounds are only semi-robust, with cut-offs in the local Péclet and Damköhler numbers.
  - **Local adaptive DG** (arXiv 2004.07148, 2020): uses the augmented norm, but its authors state that they could not prove efficiency for the new estimator.
  - **Verfürth 2005; Cai & Cai, Numer. Math. 2025:** efficiency is global in the augmented (dual) norm, with parameter-independent constants.
  - **The maximum-norm counterpart** (arXiv 2207.08251; IMA J. Numer. Anal. 43 (2023)): a weighted convective seminorm plays the role of the dual norm. Without it, the natural estimator is not efficient.
- **The search did not find** a combination of local efficiency, a localised dual norm and full robustness for convection–diffusion, as of 2026 (abstract-level).

B2. Frontier additions

result	scheme	estimate	hypotheses	mechanism	uniform in
Local adaptive DG, arXiv 2004.07148 (2020)	DG, CDR	reliable in the augmented norm; **efficiency not proved**	—	augmented norm	—
Maximum-norm estimates, arXiv 2207.08251 / IMA J. Numer. Anal. 43 (2023)	conforming, convection–diffusion	max-norm, with a weighted convective seminorm	—	Green's function	ε (with the seminorm)
Cai & Cai, Numer. Math. (2025)	conforming, CDR	hybrid, reliable and efficient, parameter-independent	global efficiency	Verfürth's dual norm	ε, reaction

B3. Open items, ranked by novelty-to-effort

1. **U1:** text only.
2. **The cellwise-σ track** (removes c_b, T5's restriction and U1's term). Medium effort; the most useful extension.
3. **A5:** read Licht 2019, Theorem 1. **EARM/Li:** read the full texts.
4. **The stopped iterate against the discrete solution.** Open, high risk.

B4. References (new in round 6)

- A posteriori analysis of a local adaptive DG method for CDR, arXiv 2004.07148. [abstract]
- Maximum norm a posteriori error estimates for convection–diffusion problems, arXiv 2207.08251, IMA J. Numer. Anal. 43(5). [abstract]
- D. Cai, Z. Cai, Numer. Math. (2025). [arXiv version read earlier]
- A. Ern, A. F. Stephansen, M. Vohralík, JCAM 234 (2010). [read]
- M. Vohralík, residual flux-based estimates for FV, Numer. Math. (2008). [abstract]
- M. Eigel, C. Merdon, J. Sci. Comput. 67 (2016). [read]